



CareLink

AI-Powered Blood Pressure Monitor

User Manual

Version 1.0

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1. Product Overview

CareLink is a smartphone application designed to make blood pressure monitoring easy, accurate, and worry-free for seniors and patients with limited technical experience. Instead of asking users to manually read and record numbers from a blood pressure cuff, CareLink's built-in AI camera simply looks at the monitor screen and reads the numbers automatically — no typing required.

The Problem It Solves

Many older adults struggle to accurately read small numbers on blood pressure monitors and record them into a notebook or phone. Misread values, forgotten readings, and lack of clinician visibility are common problems that can lead to undetected hypertension. CareLink eliminates this friction entirely.

Who It Is Designed For

- Seniors (65+) with limited smartphone experience
- Patients managing hypertension or cardiovascular conditions
- Caregivers who assist elderly family members with health monitoring
- Clinicians who need remote visibility into patient BP trends

Primary Features

Feature	Description
AI Camera Reading	Point the phone at the BP monitor; Claude AI reads the numbers automatically
Voice Guidance	Step-by-step spoken instructions walk users through the entire measurement process
Auto Photo Capture	The app takes the photo automatically (10-second countdown) if the user forgets to tap
AI Voice Assistant	Ask health questions by voice and receive spoken answers
History & Trends	All readings saved and grouped by day; daily averages displayed at a glance
Cloud Sync	Readings securely uploaded to the clinician dashboard in real time

2. System Requirements

Patient iOS App

Requirement	Specification
Operating System	iOS 15.0 or later
Device	iPhone 8 or later (physical device required — camera & microphone)
Storage	Minimum 100 MB free space
Internet	Wi-Fi or cellular data required for AI analysis and cloud sync
Camera	Rear or front camera (auto-selected by user)
Microphone	Required for voice input and voice guidance
Claude API Key	Anthropic account with active API access
OpenAI API Key	OpenAI account with Whisper API access (speech-to-text)
Backend URL	Network access to the CareLink REST API server

Clinician Dashboard (Web)

Requirement	Specification
Browser	Chrome 110+, Safari 16+, Firefox 110+, or Edge 110+
Internet	Broadband connection required
Backend	CareLink Node.js/Python API server running and accessible

Development Environment (for building from source)

Requirement	Specification
macOS	macOS 13.0 (Ventura) or later
Xcode	Xcode 15.0 or later
Swift	Swift 5.9 (included with Xcode 15)
CocoaPods / SPM	Not required — no third-party dependencies
Apple Developer Account	Free account sufficient for personal device testing

3. Installation & Setup

Follow these steps carefully. No prior programming experience is needed to run the app on your phone. The build steps require a Mac.

1

Accept the GitHub Assignment

Click the assignment link provided by your instructor. This creates a private GitHub repository for your team. Accept the invitation and note the repository URL.

2

Clone the Repository

Open Terminal (Finder → Applications → Utilities → Terminal) and run: `git clone https://github.com/[your-org]/[repo-name].git` `cd [repo-name]`

3

Open the Project in Xcode

Double-click `carelink.xcodeproj`, or in Terminal run: `open carelink.xcodeproj` Xcode will open. Wait for indexing to complete (progress bar in top center).

4

Connect Your iPhone

Plug your iPhone into your Mac with a USB cable. If prompted on iPhone, tap 'Trust'. In Xcode, click the device selector (top bar, left of the play button) and choose your iPhone.

5

Set the Development Team

In Xcode: click the project name (`carelink`) in the left panel → select the `carelink` target → go to Signing & Capabilities → choose your Apple ID under Team. If you don't see your Apple ID, go to Xcode → Settings → Accounts → add your Apple ID.

6

Build and Install

Press `⌘ + R` (or click the triangular Play button). Xcode compiles the app and installs it on your phone. The first build takes 1–3 minutes.

7

Trust the Developer Certificate on iPhone

On iPhone: Settings → General → VPN & Device Management → your Apple ID → Trust. This one-time step is required for apps not distributed through the App Store.

8

Configure API Keys (in-app)

Open CareLink on your phone → tap Settings (bottom right tab). • Enter your Claude API Key (from `console.anthropic.com`). • Enter your OpenAI API Key (from `platform.openai.com`) for voice input. • Enter the Backend API URL (ask your team's backend developer for the IP address).

Grant Permissions

The first time you tap Measure or the microphone, iOS will ask for Camera and Microphone permission. Tap Allow for both. These permissions can be reviewed later in iPhone Settings → CareLink.

4. Operating Instructions

Starting the App

Tap the CareLink icon on your iPhone home screen. The app opens to the Home tab (voice assistant). Ensure your volume is turned up — the app speaks aloud throughout.

Taking a Blood Pressure Measurement

- 1** **Open the Measure tab**
Tap the heart icon at the bottom center of the screen labeled Measure.
- 2** **Tap Start Measurement**
The app begins voice guidance immediately. Listen carefully and follow each spoken instruction.
- 3** **Step 1 — Turn on your BP monitor**
Power on your blood pressure device.
- 4** **Step 2 — Put on the cuff**
Wrap the cuff around your upper arm as instructed by your device's manual.
- 5** **Step 3 — Press START on the monitor**
Press the START button on your blood pressure device.
- 6** **Step 4 — Wait for the measurement**
Sit still for approximately 60 seconds while the cuff inflates and deflates. The app will announce progress at the 30-second mark.
- 7** **Step 5 — Point camera at screen**
When numbers appear on the monitor display, hold your phone camera toward the screen. The large pink Take Photo button will pulse. Either tap it OR wait 10 seconds — the app takes the photo automatically.
- 8** **Review the result**
The app shows the detected systolic, diastolic, and pulse values. Tap Save if correct, Re-capture to try again, or Edit to type values manually.
- 9** **Reading is saved**
The reading is saved locally and uploaded to the clinician dashboard automatically.

Using the AI Voice Assistant

Tap the Home tab → tap the microphone button → speak your question (e.g., "Is 130 over 85 high?"). The app transcribes your speech, sends it to the AI, and reads the answer aloud. The microphone disables itself during analysis to prevent accidental double-taps. It re-enables automatically when the response is ready.

Viewing History

Tap the History tab. Readings are grouped by date. If you took 2 or more readings in one day, an Avg row appears at the bottom of that day showing the average values. Tap any individual row to see details. A tip at the top encourages 5 readings per day for better accuracy.

Shutting Down

CareLink does not require a formal shutdown. Simply press the iPhone home button or swipe up to return to the home screen. All data is saved automatically. Pending cloud uploads complete in the background.

5. Technical Specifications

Architecture

CareLink follows an MVC (Model-View-Controller) pattern implemented in Swift/UIKit. The app communicates with external AI APIs over HTTPS and syncs data to a REST backend. All local data is persisted using UserDefaults and in-memory arrays. There is no local database — readings are stored in a JSON-serializable array keyed in UserDefaults and mirrored to Firestore via the backend.

Technologies Used

Layer	Technology	Purpose
UI	Swift 5.9 / UIKit	Native iOS interface, all screens
Camera	AVFoundation (AVCaptureSession)	Live camera feed, photo capture, video frames
Speech Output	AVSpeechSynthesizer	Text-to-speech for voice guidance
Speech Input	AVAudioRecorder + OpenAI Whisper	Record audio; transcribe to text
AI Vision	Claude claude-opus-4-6 (Anthropic)	Read BP numbers from photos
AI Chat	Claude claude-sonnet-4-6 (Anthropic)	Answer health questions
AI Guidance	Claude claude-haiku-4-5-20251001	Behavior-based measurement tips
Image Processing	Core Image (CIColorControls)	Enhance contrast before sending to AI
Networking	URLSession	HTTPS API calls to Anthropic, OpenAI, backend
Local Storage	UserDefaults	API keys, settings, readings cache
Cloud	REST API → Firestore	Sync readings to clinician dashboard

AI Model Configuration

Task	Model	Max Tokens	Timeout
BP Image Reading	claude-opus-4-6	120	50 s
AI Chat	claude-sonnet-4-6	500	60 s
Behavior Guidance	claude-haiku-4-5-20251001	56	25 s
Speech-to-Text	whisper-1 (OpenAI)	—	30 s

Data Flow

1. User completes measurement → 2. Photo captured by AVCapturePhotoOutput → 3. Image resized (max 1536 px) and contrast-enhanced (Core Image) → 4. Base64-encoded and sent to Claude Vision API → 5. JSON response parsed (systolic, diastolic, pulse) → 6. User confirms → 7. BloodPressureReading saved to UserDefaults → 8. CloudSyncService POSTs reading to backend REST API → 9. Backend writes to Firestore under patient document.

Firestore Patient Schema

Field	Type	Example
id	string	P-2025-001
firstName / lastName	string	Robert / Anderson
dateOfBirth	string (ISO 8601)	1958-03-15
diagnosis	array of strings	['Hypertension', 'Type 2 Diabetes']
medications	array of maps	{name, dosage, frequency, startDate}
targetSystolic / targetDiastolic	number	130 / 80
riskLevel	string	high
assignedClinicianId	string	clinician-001
address	map	{street, city, state, zipCode}
emergencyContact	map	{name, phone, relationship}

6. Data Privacy, Security & Known Limitations

Data Privacy

- Camera images taken for BP reading are transmitted directly to the Anthropic API over TLS/HTTPS and are NOT stored on Anthropic servers beyond the API call duration per Anthropic's data usage policy.
- Voice recordings are sent to OpenAI's Whisper API over HTTPS and are subject to OpenAI's privacy policy. Audio is not stored locally after transcription.
- Blood pressure readings and patient identifiers are transmitted to the backend server. Ensure the backend is hosted on a HIPAA-compliant infrastructure if used in a clinical setting.
- API keys are stored in iOS UserDefaults (device local storage). Keys are not transmitted anywhere other than the respective API endpoints.
- No analytics, crash reporting, or third-party tracking SDKs are included in this application.

Security Warnings

- Never share your Claude or OpenAI API keys. Anyone with your key can incur charges on your account.
- The backend API URL should use HTTPS in production. Using HTTP (e.g., <http://192.168.x.x>) is acceptable only on a private LAN during development.
- Patient health data (BP readings) should only be stored on HIPAA-compliant servers in a real clinical deployment.
- Lock your iPhone with a passcode to prevent unauthorized access to stored readings and API keys.

Known Operational Limitations

Limitation	Details / Workaround
Camera required	The app cannot run on the iOS Simulator. A physical iPhone is mandatory.
AI reading accuracy	Claude Vision performs best with clear, well-lit monitor screens. Poor lighting, glare, or blurry
Whisper language	Voice input defaults to auto-detect. Performance is best in English. Non-English languages m
Silence detection	Auto-stop for voice recording triggers at -40 dB after 2 seconds of silence. Very quiet environ
Offline mode	AI features (camera reading, voice chat) require an internet connection. Readings can be sav
BP monitor compatibility	The AI reads digital display screens. Analog (dial) manometers are not supported.
API rate limits	Heavy usage may hit Anthropic or OpenAI rate limits. If errors occur, wait 30 seconds and ret

7. Troubleshooting

Symptom	Likely Cause	Solution
"API key not configured"	No API key entered in Settings	Settings tab → enter Claude API key
Camera shows black screen	Running on Simulator or permissions not granted on iPhone	Use real iPhone; grant camera permission in iOS Settings
AI cannot read numbers	Poor lighting, blurry image, or small display	Hold phone steady, move closer, ensure good lighting; try larger display
"Model error" in chat	Incorrect or expired API key	Settings → re-enter a valid Claude API key
Voice reads emoji aloud	Should not occur (fixed)	Update to latest version — emoji are stripped before TTS
Cloud sync fails	Wrong backend URL or server offline	Settings → verify backend URL; check server is running
Mic button unresponsive	Analysis still in progress	Wait for the current response to finish — mic re-enables automatically
Auto photo not firing	User stopped the measurement flow	Re-tap Start Measurement and proceed through all 5 steps

CareLink v1.0 • User Manual • Built for seniors who need simple, reliable blood pressure monitoring.